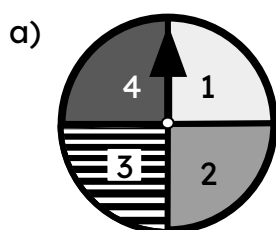




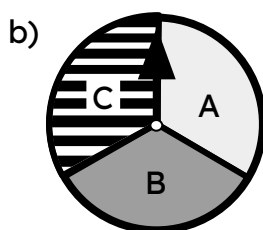
Equally likely outcomes

Task A: Listing possible outcomes

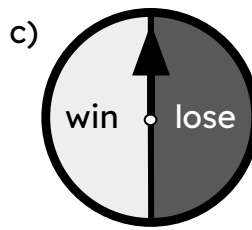
1) In each question a **trial** consists of a single spin of the spinner. Write down the **sample space** for each **trial**.



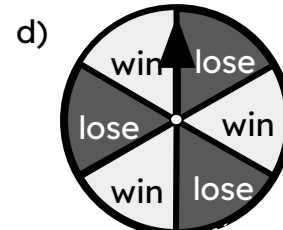
$$\xi = \{1, 2, 3, 4\}$$



$$\xi = \{A, B, C\}$$

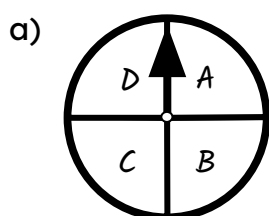


$$\xi = \{\text{win}, \text{lose}\}$$



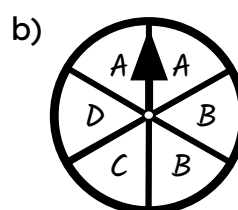
$$\xi = \{\text{win}, \text{lose}\}$$

2) In each question a **trial** consists of a single spin of the spinner. Use the sample spaces to complete the spinner. There may be more than one possible answer.



$$\xi = \{A, B, C, D\}$$

Note: the letters may go in any order.

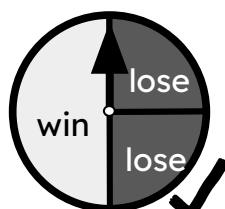
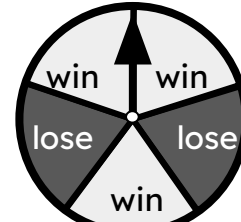
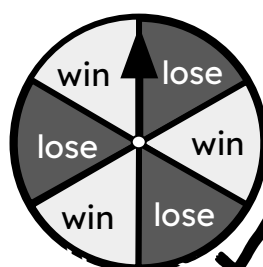
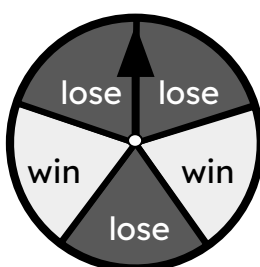


$$\xi = \{A, B, C, D\}$$

Note: any of the letters may be repeated any number of times, so long as all four letters are present.

Task B: Recognising when two outcomes are equally likely

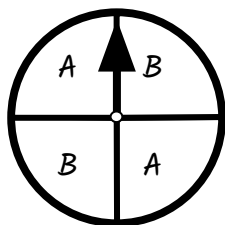
1) Tick the spinners where the **outcomes** 'win' and 'lose' are equally likely.



Task C: Recognising more than two equally likely outcomes

1) Complete the spinners to make the outcomes equally likely to happen.

a) $\xi = \{A, B\}$



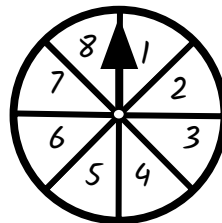
two sectors for each A and B

b) $\xi = \{\text{win, lose}\}$



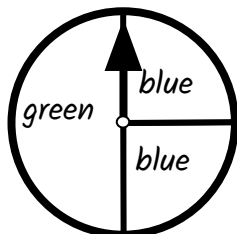
three sectors for each win and lose

c) $\xi = \{\text{even number, odd number}\}$



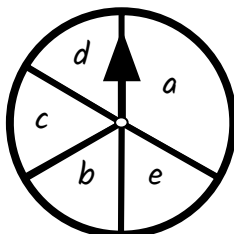
any four even numbers and any four odd numbers
numbers may be repeated

d) $\xi = \{\text{green, blue}\}$



colours may be switched

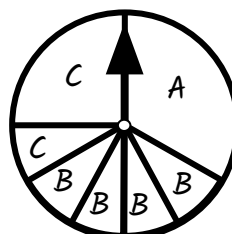
e) $\xi = \{\text{vowel, consonant}\}$



letter may be repeated

The large sector is the same size as two small sectors.

f) $\xi = \{A, B, C\}$



The largest sector is the same size as four lots of the smallest sector.

The next largest is the same size as three lots.

2) Aisha, Jun and Izzy are playing a game that involves rolling a regular 6-sided dice.

Aisha's piece moves if the dice roll is 1 or 2.

Jun's piece moves if the dice roll is 3 or 4.

Izzy's piece moves if the dice roll is 5 or 6.

a) Izzy thinks that 6 is the most difficult number on the dice to roll.

Explain why this is incorrect.

Each number on the dice is equally likely to be rolled.

b) The **sample space** to show the possible **outcomes** to each **trial**.

$\xi = \{\text{Aisha moves, Jun moves, Izzy moves}\}$

Explain why these outcomes are equally likely to happen.

Each player has the same number of outcomes which move their piece and each outcome (number on the dice) is equally likely to be rolled.

